cTheorems

sahasatvik
https://github.com/sahasatvik/typst-theorems

Contents

This package provides functions that help create numbered theorem environments, taking heavy inspiration from the LaTeX packages `amsthm`, `thmtools`, `apxproof`.

A *theorem environment* combines content with automatically updated *numbering* information. Theorem environments can

- share the same counter (*Theorems* and *Lemmas* often do so)
- have their counters attached to headings or other environments (*Corollaries* are often numbered based upon the parent *Theorem*)
- be `<label>`-ed and `@reference`-d
- be restated or deferred to appear later in the document.

This package also introduces a few miscellaneous features related to mathematical writing.

- Proof environments, with *QED* symbols.
- Equation tags.
- Predefined sets of commonly used theorem environments, and a few themes.

1. Setup

```
#import "@preview/ctheorems:2.0.0": *
#show: thm-rules     // Must include!
```

*Applying the `thm-rules` show rule — in the same `.typ` file where `ctheorems` functions have been used — is **essential** for displaying theorem environments, references, and equations correctly!*

A standard set of theorem environments in the AMS style is available using

```
#import thm-themes.ams: *
```

This lets you immediately use `theorem`, `proof`, `proposition`, `lemma`, `corollary`, `definition`, `example`, `remark`, `claim`. See Appendix B.1 for more details.

2. Demonstration

```
#import "@preview/ctheorems:2.0.0": *
#import thm-themes.ams: *
#show: thm-rules.with(
  qed-symbol: $square$
)

#definition[Expectation][
  The expectation of a random variable $X$ on
a probability space $(Omega, cal(E), PP)$ is
$
    EE[X] = integral X dif PP,
  $ whenever well-defined.
] <expectation>

#remark[
  We require at least one of $EE[X^+],
EE[X^-]$ to be finite.
]

#proposition[
  For any $A in cal(E)$, $
    PP(X in A) = EE[bold(1)_A (X)].
  $
] <prob-exp>

#theorem[Markov][
  Let $X ≥ 0$. For all $a > 0$, $
    PP(X > a) ≤ EE[X] / a.
  $
] <markov>

#corollary[Chebyshev][
  Let $EE[X] = mu$, $"var"[X] = sigma^2$.
Then, $
    PP(|X - mu| ≥ k sigma) ≤ 1 / k^2.
  $
] <chebyshev>

#proof[of @markov][
  $
    PP(X &> a) \
      &= EE[bold(1)_((a, oo))(X)]
        #tag[(@prob-exp[Prop.])] \
      &≤ EE[(X / a) bold(1)_((a, oo))(X)] \
      &≤ EE[X] / a. #qedhere
  $
]
```

Definition 2.1 (Expectation)**.** The expectation of a random variable X on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ is

$$\mathbb{E}[X] = \int X \, \mathrm{d}\mathbb{P},$$

whenever well-defined.

Remark. We require at least one of $\mathbb{E}[X^+], \mathbb{E}[X^-]$ to be finite.

Proposition 2.2. *For any $A \in \mathcal{E}$,*

$$\mathbb{P}(X \in A) = \mathbb{E}[\mathbf{1}_A(X)].$$

Theorem 2.3 (Markov)**.** *Let $X \geq 0$. For all $a > 0$,*

$$\mathbb{P}(X > a) \leq \frac{\mathbb{E}[X]}{a}.$$

Corollary 2.3.1 (Chebyshev)**.** *Let $\mathbb{E}[X] = \mu$, $\mathrm{var}[X] = \sigma^2$. Then,*

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Proof of Theorem 2.3.

$$\begin{aligned}
&\mathbb{P}(X > a) \\
&\quad = \mathbb{E}\left[\mathbf{1}_{(a,\infty)}(X)\right] \quad \text{(Prop. 2.2)} \\
&\quad \leq \mathbb{E}\left[\left(\frac{X}{a}\right)\mathbf{1}_{(a,\infty)}(X)\right] \\
&\quad \leq \frac{\mathbb{E}[X]}{a}.
\end{aligned}$$

□

3. Features

3.1. Numbered theorems

```
#let theorem = thm.with(
  supplement: "Theorem",
  base-level: 1
)

#theorem("Euclid")[
  There are infinitely many primes.
] <euclid>
```

Theorem 3.1 (Euclid)**.** *There are infinitely many primes.*

Note that `theorem` inherits its numbering from the current heading (the default `thm.base`). By setting `thm.base-level` to `1`, this theorem only uses the first level count from its base.

3.2. References

Since our theorem has been labeled `<euclid>`, we can reference it elsewhere in the document via `@euclid`.

```
We will supply a proof of @euclid later.
```

We will supply a proof of Theorem 3.1 later.

3.3. Chained numbering

Theorem environments can be attached together by setting the child's `thm.base` to the parent's `thm.counter`.

```
#let corollary = thm.with(
  supplement: "Corollary",
  base: "Theorem"
)

#corollary(restate: true)[
  There is no largest prime number.
] <largest-prime>

#proof([of @largest-prime], defer: true)[
  The existence of a largest prime would
imply the finitude of the set of primes.
]
```

Corollary 3.1.1. *There is no largest prime number.*

We have set `thm.restate` and `thm.defer` — go to Section 3.7 to find out more!

3.4. Shared numbering

Theorem environments can share the same numbering by setting a common `thm.counter`.

```
#let proposition = thm.with(
  supplement: "Proposition",
  counter: "Theorem",
  base-level: 1
)

#proposition[
  If a natural number divides $a$ and $b$, it
also divides $a - b$.
]
```

Proposition 3.2. *If a natural number divides a and b, it also divides $a - b$.*

For directly manipulating theorem counters, see `thm-counter-get`, `thm-counter-step`, `thm-counter-update`.

3.5. Proofs

Theorems go hand in hand with proofs, which are available using `proof`.

```
#proof[of @euclid][
  Suppose to the contrary that $p_1, p_2,
  dots, p_n$ is a finite enumeration of
  all primes. Set $P = p_1 p_2 dots p_n$.
  Since $P + 1$ is not in our list, it cannot
  be prime. Thus, some prime factor $p_j$
  divides $P + 1$. Since $p_j$ also divides
  $P$, it must divide the difference $(P + 1)
  - P = 1$, a contradiction.
]
```

Proof of Theorem 3.1. Suppose to the contrary that $p_1, p_2, \ldots, p_n$ is a finite enumeration of all primes. Set $P = p_1 p_2 \ldots p_n$. Since $P + 1$ is not in our list, it cannot be prime. Thus, some prime factor p_j divides $P + 1$. Since p_j also divides P, it must divide the difference $(P + 1) - P = 1$, a contradiction. ■

Proof environments will float a *QED* symbol (■) at the bottom right by default. The symbol can be customized by setting `thm-rules.qed-symbol`. If your proof ends in a block equation, or a list/enum, you can place `qedhere` to correctly position the *QED* symbol.

```
#show: thm-rules.with(qed-symbol: $square$)

#theorem[Prime gaps][
  There are arbitrarily long stretches of
composite numbers.
]
#proof[
  For any $n > 2$, consider $
    n! + 2, quad
    n! + 3, quad ... , quad
    n! + n. #qedhere
  $
]
```

Theorem 3.3 (Prime gaps)**.** *There are arbitrarily long stretches of composite numbers.*

Proof. For any $n > 2$, consider

$$n! + 2, \quad n! + 3, \quad \ldots, \quad n! + n. \qquad \square$$

Caution: `qedhere` does not play well with numbered equations!

3.6. Equation tags

Tags can be inserted into equations using `tag`. These float to the right, mimicking `\tag` in LaTeX.

```
$
  (a + b)^2
    &= a^2 + 2 a b + b^2 \
    &≤ 2a^2 + 2b^2 #tag[(AM-GM)] \
    &≤ 2 thin max{a, b}^2.
$
```

$$\begin{aligned}(a+b)^2 &= a^2 + 2ab + b^2 \\ &\leq 2a^2 + 2b^2 \quad \text{(AM-GM)} \\ &\leq 2\max\{a,b\}^2.\end{aligned}$$

3.7. Defer, restate, and display

Theorem environments can be restated, or deferred to later in the document by setting `thm.restate`, `thm.defer` as in the example in Section 3.3, then calling `thm-restate`.

```
#import thm-state: thm-restate, thm-display

#thm-restate()
```

Corollary 3.1.1. *There is no largest prime number.*

Proof of Corollary 3.1.1. The existence of a largest prime would imply the finitude of the set of primes. ■

This is often useful for pushing content to the appendix. For finer control, consider using `thm.restate-keys`.

```
Euclid's Theorem states:
#thm-restate(
  "Euclid",
  all: true,
  fmt: thm => thm.body
)
```

Euclid's Theorem states: There are infinitely many primes.

The `thm-display` function is the most powerful method for manipulating theorem environments. The following produces a crude outline of named theorem environments (so far), excluding proofs; see `thm.fmt` for more information about the formatting function `fmt`.

```
#thm-display(
  thm => thm.name != none and thm.supplement != "Proof",     // Filter
  fmt: thm => {                                              // Format
    let head =[*#thm.supplement~#thm.number*]
    if thm.name != none { head = head + [~(#thm.name)] }
    let page = link(thm.loc, [#thm.loc.position().page])     // Get page number
    [#head~#box(width: 1fr, repeat[.])~#page\ ]
  }
)
```


Setting `thm-display.final` to `true` would extend this search to the entire document.

3.8. Formatting

The default theorem formatting function, `thm-fmt-block` helps with basic customization of the theorem title and body, along with the surrounding block.

```
#let theorem-standout = theorem.with(
  stroke: 1pt + green,
  fill: green.lighten(95%),
  outset: 0.7em,
  spacing: 1.5em
)

#theorem-standout(
  title-fmt: x => smallcaps(strong(x))
)[
  There are infinitely many composite
numbers.
]
```

Theorem 3.4. *There are infinitely many composite numbers.*

See `thm.fmt` for more information on how to write your own custom formatting function.

4. Acknowledgements

Thanks to

- MJHutchinson for suggesting and implementing the `base-level` and `base: none` features,

- rmolinari for suggesting and implementing the `separator: ...` feature,
- DVDTSB for contributing
  - the idea of passing named arguments from the theorem directly to the `fmt` function.
  - the `number: ...` override feature.
  - the `title: ...` override feature in `thm-plain`.
- PgBiel for fixing breaking changes in version updates.
- The authors of the LaTeX packages `amsthm`, `thmtools`, `apxproof`.
- The awesome devs of typst.app for their support.

A. Function documentation

A.1. ctheorems

- thm()
- proof()
- tag()
- thm-rules()
- thm-fmt-block()
- proof-body-fmt()

Variables:

- qedhere

thm

Numbered theorem environment.

```
#thm(base: none)[Name][
  #lorem(10)
]

#thm(supplement: "Corollary", base: "Theorem")[
  #lorem(7)
]
```

Theorem 1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.*

Corollary 1.1. *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Parameters

```
thm(
  ..args: arguments,
  body: content,
  supplement: str content,
  counter: str auto,
  base: str none,
  base-level: int none,
  name: content,
  number: content auto none,
  numbering: str function none,
  restate: bool,
  defer: bool,
  restate-keys: array,
  fmt: function,
  ref-fmt: function
) -> figure context
```

..args `arguments`

Extra arguments. See `thm.name`.

body `content`

Theorem body.

supplement `str` or `content`

Theorem supplement.

Default: `"Theorem"`

counter `str` or `auto`

Environment counter name. If `auto`, defaults to `thm.supplement`.

```
#thm(
  supplement: "Proposition",
  counter: "Theorem",
  base: none
)[
  #lorem(7)
]
```

Proposition 2. *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Default: `auto`

base `str` or `none`

Base counter name, whose numbering prefixes the theorem environment numbering. If `"heading"`, use the heading counter. If `none`, the theorem environment maintains a global count with no prefix.

```
#thm(supplement: "Example", base: "Theorem")[
  #lorem(4)
]
```

Example 2.1. *Lorem ipsum dolor sit.*

Default: `"heading"`

base-level `int` or `none`

Base level, determining the number of levels of the `base` numbering to use for the theorem environment numbering. If `none`, all levels from the `base` numbering are used. Setting a base level higher than what the `base` provides will introduce zeros for padding.

```
#thm(
  supplement: "Sub-Sub-Theorem",
  base: "Theorem",
  base-level: 2
)[
  #lorem(12)
]
```

Sub-Sub-Theorem 2.0.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor.*

Default: `none`

name `content`

Theorem name. If `auto`, the first positional argument from `thm.args` (if present) is chosen as the theorem environment's name.

```
#thm(name: "Named", base: none)[
  #lorem(4)
]
#thm(base: none)[Also Named][
  #lorem(8)
]
```

Theorem 3 (Named)**.** *Lorem ipsum dolor sit.*

Theorem 4 (Also Named)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Default: `auto`

number `content` or `auto` or `none`

Theorem number. If `auto` (and `thm.numbering` is not `none`), the number is computed based on `thm.counter`, `thm.base`, `thm.base-level`.

```
#thm(number: $dagger dagger$, base: none)[
  #lorem(5)
] <d-dag>

Recall @d-dag.
```

Theorem ††. *Lorem ipsum dolor sit amet.*

Recall Theorem ††.

Default: `auto`

numbering `str` or `function` or `none`

Theorem numbering style. If `none`, numbering is suppressed.

```
#thm(supplement: "Remark", numbering: none)[
  #lorem(5)
]
```

Remark. *Lorem ipsum dolor sit amet.*

Default: `"1.1"`

restate `bool`

Mark for being restated later. See `thm-restate()`.

Default: `false`

defer `bool`

Mark for being deferred to later, without appearing in the current position. See `thm-restate()`.

Default: `false`

restate-keys `array`

Keys used by `thm-restate.keys` for filtering. If `auto`, defaults to an array containing `thm.supplement` and `thm.name` (if present).

Default: `auto`

fmt `function`

Formatting function controlling the appearance of the theorem environment, of the form `thm` ⇒ `content`. `thm` is a dictionary whose keys are essentially the arguments of `thm()`, with a couple of additions/exceptions.

- The keys `name`, `counter`, `number`, `restate-keys` contain their finally inferred/computed values.
- The key `loc` contains the theorem's location.
- The key `args` contains a dictionary of only the named arguments from `thm.args`.

It is often useful to use `thm.args` to control formatting functions.

```
#let thm-color = thm.with(
  base: none,
  fmt: thm => {
    let color = thm.args.at("color", default:
black)
    [*#thm.supplement~#thm.number.* #text(color,
thm.body)]
  }
)

#thm-color[#lorem(15)]
#thm-color(color: blue)[#lorem(7)]
```

Theorem 5. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Theorem 6. Lorem ipsum dolor sit amet, consectetur adipiscing.

Default: `thm-fmt-block`

ref-fmt `function`

Formatting function for references, with the same form as `thm.fmt`. The `thm` dictionary carries an additional `ref-supplement` key containing the supplement used in the `@cite[ref-supplement]` call.

```
#let thm-ref-name = thm.with(
  base: none,
  ref-fmt: thm => {
    [#thm.name~(#thm.supplement~#link(thm.loc,
thm.number))]
  }
)

#thm-ref-name[Gauss][#lorem(17)] <gauss>

By @gauss, ...
```

Theorem 7 (Gauss)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore.*

By Gauss (Theorem 7), ...

Default:
```
thm => {
    let supplement = thm.supplement
    if thm.ref-supplement ≠ none { supplement = thm.ref-supplement }
    if supplement ≠ none and supplement ≠ [] { supplement = [#supplement~] }
    [#supplement#link(thm.loc, thm.number)]
  }
```

proof

Proof environment. Identical to `thm()`, with different defaults. Uses the `thm-fmt-block()` formatting function (for `thm.fmt`), with `proof-body-fmt()` (for `thm-fmt-block.body-fmt`).

```
#thm(base: none)[
  #lorem(6)
]
#proof[
  #lorem(3)
]
```

Theorem 8. *Lorem ipsum dolor sit amet, consectetur.*

Proof. Lorem ipsum dolor. ■

Parameters

`proof() -> function`

tag

Add content which floats to the right in an equation.

```
$
  (a + b)^2
    &= a^2 + 2 a b + b^2 \
    &≤ 2a^2 + 2b^2 #tag[(AM-GM)] \
    &≤ 2 thin max{a, b}^2.
$
```

$$
\begin{aligned}
(a + b)^2 &= a^2 + 2ab + b^2 \\
&\leq 2a^2 + 2b^2 \qquad \text{(AM-GM)} \\
&\leq 2 \max\{a, b\}^2.
\end{aligned}
$$

Parameters

```
tag(
  t: any,
  move: bool
) -> content
```

t `any`

Content to use as tag

move `bool`

If `false`, tags occupy no horizontal space in the equation layout. Set this to `true` if equation content overlaps the tag. This will have the minor disadvantage of moving the equation off-center to accommodate the tag.

Note: Trying to determine this option automatically currently causes layout convergence issues in some cases.

Default: `false`

thm-rules

Rules for styling theorem environments, references, proofs, etc. *Must appear at the beginning of the document.*

Parameters

```
thm-rules(
  qed-symbol: content,
  doc
) -> content
```

qed-symbol `content`

Symbol displayed at the end of proofs. See `proof()`, `qedhere`, `proof-body-fmt()`. Use as

```
#show: thm-rules.with(
  qed-symbol: $square$
)

#proof[
  #lorem(5)
  $ integral_0^oo sin(x)/x thin d x = pi/2.
#qedhere $
]
```

Proof. Lorem ipsum dolor sit amet.

$$\int_0^\infty \frac{\sin(x)}{x}\, dx = \frac{\pi}{2}. \qquad \square$$

Default: `$qed$`

thm-fmt-block

Default formatting function for theorem environments, wrapping everything in a block. Intended for use in `thm.fmt`. Named arguments from `thm.args` can override all of the arguments below; any remaining named arguments are passed to the block. Blocks have `width: 100%` set by default.

```
#thm(
  base: none,
  fmt: thm-fmt-block,
  title-fmt: x => strong(smallcaps(x)),
  separator: [\ ],
  stroke: (left: 1pt),
  inset: (x: 0.7em, y: 0.2em)
)[Special][
  #lorem(16)
]
```

Theorem 9 (Special)
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

Parameters

```
thm-fmt-block(
  thm: dictionary,
  block-args: dictionary,
  name-fmt: function,
  title-fmt: function,
  body-fmt: function,
  separator: content
) -> content
```

thm `dictionary`

Theorem data. See `thm.fmt` for details.

block-args `dictionary`

Named arguments for the block.

Default: `(:)`

name-fmt `function`

Formatting for the environment name.

Default: `x => [(#x)]`

title-fmt `function`

Formatting for the environment title (head and number).

Default: `strong`

body-fmt `function`

Formatting for the environment body.

Default: `emph`

separator `content`

Separator between title and body.

Default: `[*.* ]`

proof-body-fmt

Used as `thm-fmt-block.body-fmt` by `proof()`, for properly styling proofs with a `qed` symbol inserted at the end of the body. Also see `qedhere`.

```
#show: thm-rules.with(qed-symbol: $"QED"$)

#proof-body-fmt[#lorem(3)]

#proof-body-fmt[
  $
    phi.alt(x) = 1/sqrt(2 pi) e^(-x^2\/2) #qedhere
  $
]
```

Lorem ipsum dolor. QED

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \text{QED}$$

Parameters

`proof-body-fmt(body: content) -> content`

body `content`

Proof body.

qedhere `metadata`

If placed in a block equation/enum/list within a proof, place a qed symbol to its right.

```
#proof[
  #lorem(3)
  $ x^2 + y^2 = z^2. #qedhere $
]

#proof[
  + #lorem(4)
  + #lorem(5) #qedhere
]

#proof[
  $
    (a + b)^2 &= (a + b)(a + b) \
              &= a^2 + 2 a b + b^2. #qedhere
  $
]
```

Proof. Lorem ipsum dolor.

$$x^2 + y^2 = z^2. \qquad \blacksquare$$

Proof.

1. Lorem ipsum dolor sit.
2. Lorem ipsum dolor sit amet. ■

Proof.

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b) \\ &= a^2 + 2ab + b^2. \qquad \blacksquare\end{aligned}$$

A.2. thm-state

Methods for restating, reformatting, filtering, and manipulating theorem environments from elsewhere in the document. Every call to `thm` appends a `thm` data dictionary to the state `thm-stored`, where the keys of `thm` are as described in `thm.fmt`.

Import all functions using

```
#import thm-state: *
```

- thm-restate()
- thm-display()

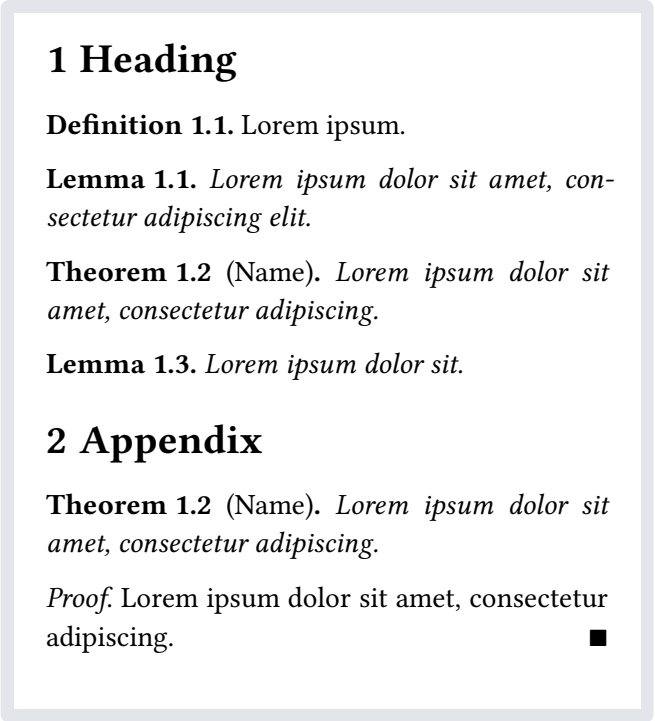
Variables:

- thm-stored

thm-restate

Displays theorem environments which have been marked to be restated or deferred (by setting `thm.restate`, `thm.defer`), can be filtered. Useful for pushing content to the appendix.

```
#let theorem = thm.with(supplement: "Theorem")
#let lemma = thm.with(
  supplement: "Lemma",
  counter: "Theorem",
)
#let definition = thm.with(
  supplement: "Definition",
  body-fmt: x => x
)

= Heading

#definition[#lorem(2)]
#lemma[#lorem(8)]

#theorem("Name", restate: true)[#lorem(7)]
#proof(defer: true)[
  #lorem(7)
]

#lemma[#lorem(4)]

= Appendix

#thm-restate()
```



1 Heading

Definition 1.1. Lorem ipsum.

Lemma 1.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Theorem 1.2 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Lemma 1.3. *Lorem ipsum dolor sit.*

2 Appendix

Theorem 1.2 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Parameters

```
thm-restate(
  ..keys: str content array function,
  fmt: function auto,
  at: label selector location function auto,
  final: bool,
  all: bool
) -> content
```

..keys `str` or `content` or `array` or `function`

String/content keys, array of keys, or functions used to filter theorem environments. A theorem environment is displayed if it passes *any* of the filters.

If `k` in `keys` is a `string`/`content`, theorem environments containing `k` in its array of `restate-keys` will be matched.

```
#let theorem = thm.with(supplement: "Theorem")
#let lemma = thm.with(
  supplement: "Lemma",
  counter: "Theorem",
)

= Heading

#theorem(restate: true)[#lorem(6)]
#lemma(restate: true)[#lorem(4)]
#proof(defer: true)[#lorem(7)]
#lemma(restate: true)[#lorem(3)]

= Restate lemmas/proofs
#thm-restate("Lemma", "Proof")
```

1 Heading

Theorem 1.1. *Lorem ipsum dolor sit amet, consectetur.*

Lemma 1.2. *Lorem ipsum dolor sit.*

Lemma 1.3. *Lorem ipsum dolor.*

2 Restate lemmas/proofs

Lemma 1.2. *Lorem ipsum dolor sit.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Lemma 1.3. *Lorem ipsum dolor.*

If `k` in `keys` is an array of `strings`/`content`, theorem environments containing *all* keys from `k` in its array of `restate-keys` will be matched.

```
= Heading

#theorem(
  "Result A",
  restate: true,
  restate-keys: ("Theorem", "Result A")
)[#lorem(6)]
#proof(
  defer: true,
  restate-keys: ("Proof", "Result A")
)[#lorem(7)]
#theorem(restate: true)[#lorem(6)]
#theorem(
  "Result B",
  restate: true,
  restate-keys: ("Theorem", "Result B")
)[#lorem(6)]
#proof(
  defer: true,
  restate-keys: ("Proof", "Result B")
)[#lorem(7)]

= Restate Result A
#thm-restate("Result A")

= Restate theorems tagged Result B
#thm-restate(("Theorem", "Result B"))
```

3 Heading

Theorem 3.1 (Result A)**.** *Lorem ipsum dolor sit amet, consectetur.*

Theorem 3.2. *Lorem ipsum dolor sit amet, consectetur.*

Theorem 3.3 (Result B)**.** *Lorem ipsum dolor sit amet, consectetur.*

4 Restate Result A

Theorem 3.1 (Result A)**.** *Lorem ipsum dolor sit amet, consectetur.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

5 Restate theorems tagged Result B

Theorem 3.3 (Result B)**.** *Lorem ipsum dolor sit amet, consectetur.*

If `k` in `keys` is a `function`, it must be of the form `restate-keys` → `bool`.

```
= Heading

#theorem(
  restate: true,
  restate-keys: (
    "Theorem", "Unproven claim"
  )
)[#lorem(6)]
#theorem(restate: true)[#lorem(6)]
#lemma(
  "Claim D",
  restate: true,
  restate-keys: ("Lemma", "Claim D")
)[#lorem(6)]

= Restate claims
#thm-restate(
  keys ⇒ keys.any(
    k ⇒ lower(k).contains("claim")
  )
)
```

6 Heading

Theorem 6.1. *Lorem ipsum dolor sit amet, consectetur.*

Theorem 6.2. *Lorem ipsum dolor sit amet, consectetur.*

Lemma 6.3 (Claim D)**.** *Lorem ipsum dolor sit amet, consectetur.*

7 Restate claims

Theorem 6.1. *Lorem ipsum dolor sit amet, consectetur.*

Lemma 6.3 (Claim D)**.** *Lorem ipsum dolor sit amet, consectetur.*

fmt `function` or `auto`

Formatting function, with the same form as `thm.fmt`. The default `auto` uses the originally supplied formatting function `thm.fmt`. See `thm-display.fmt`.

Default: `auto`

at `label` or `selector` or `location` or `function` or `auto`

Location up to which theorem environments will be displayed. The default `auto` uses the location where `thm-restate()` was called. See `thm-display.at`.

Default: `auto`

final `bool`

If `true`, display environments up to the end of the document. Overrides `thm-restate.at`. See `thm-display.final`.

Default: `false`

all `bool`

If `true`, ignores the `thm`'s `restate` or `defer` state (see `thm.restate`, `thm.defer`). Calling `thm-restate(all: true)` is equivalent to `thm-display()`.

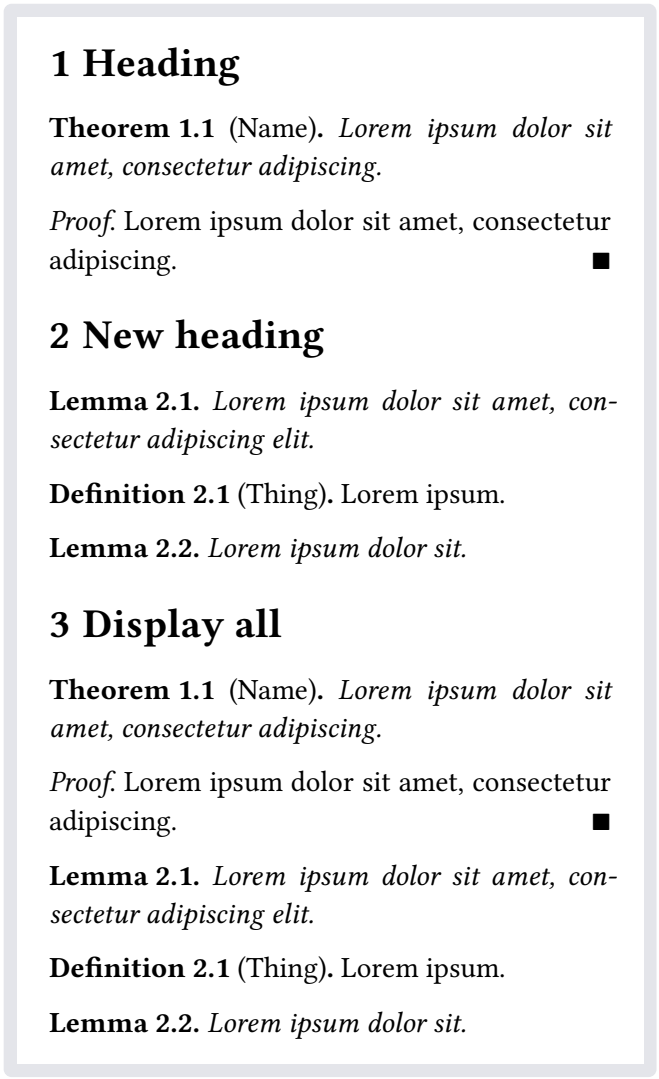
Default: `false`

thm-display

Displays all theorem environments, can be filtered.

Every call to `thm()` appends a `thm` dictionary to `thm-stored`. A `thm` dictionary contains complete information about a theorem environment; its keys are as described in `thm.fmt`. This lets theorems be restated, filtered, and otherwise manipulated elsewhere in a document.

```
#let theorem = thm.with(supplement: "Theorem")
#let lemma = thm.with(
  supplement: "Lemma",
  counter: "Theorem",
)
#let definition = thm.with(
  supplement: "Definition",
  body-fmt: x ⇒ x
)

= Heading <h1>

#theorem("Name")[#lorem(7)]
#proof[
  #lorem(7)
]

= New heading <h2>

#lemma[#lorem(8)]
#definition("Thing")[#lorem(2)]
#lemma[#lorem(4)]

= Display all

#thm-display()
```



1 Heading

Theorem 1.1 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

2 New heading

Lemma 2.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Definition 2.1 (Thing)**.** Lorem ipsum.

Lemma 2.2. *Lorem ipsum dolor sit.*

3 Display all

Theorem 1.1 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Lemma 2.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Definition 2.1 (Thing)**.** Lorem ipsum.

Lemma 2.2. *Lorem ipsum dolor sit.*

Parameters

```
thm-display(
  ..filters: function,
  fmt: function auto,
  at: label selector location function auto,
  final: bool
) -> content
```

..filters `function`

Filtering functions. Each `f` in `filters` is a function `thm` ⇒ `bool`. A theorem environment is displayed if it passes *any* of the filters.

```
= Display only theorems/proofs

#thm-display(
  thm => thm.supplement == "Theorem",
  thm => thm.supplement == "Proof",
)

= Display if `name` is present

#thm-display(
  thm => thm.name ≠ none
)
```

4 Display only theorems/ proofs

Theorem 1.1 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

5 Display if `name` is present

Theorem 1.1 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Definition 2.1 (Thing)**.** Lorem ipsum.

fmt `function` or `auto`

Formatting function, with the same form as `thm.fmt`. The default `auto` uses the originally supplied formatting function `thm.fmt`.

```
#thm-display(
  thm => thm.supplement ≠ "Proof",
  final: true,
  fmt: thm => {
    let head = [*#thm.supplement~#thm.number*]
    if thm.name ≠ none {
      head = head + [~(#thm.name)]
    }
    let page = link(thm.loc,
[#thm.loc.position().page])
    [#head~#box(width: 1fr, repeat[.])~#page\ ]
  }
)
```


Setting `final: true` ensures that even if this `thm-display()` call is placed at the beginning of the document, all theorem environments are listed.

Default: `auto`

at `label` or `selector` or `location` or `function` or `auto`

Location up to which theorem environments will be displayed. The default `auto` uses the location where `thm-display()` was called.

```
= Display up to `<h2>`

#thm-display(at: <h2>)
```

6 Display up to <h2>

Theorem 1.1 (Name)**.** *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Default: `auto`

final `bool`

If `true`, display all theorem environments up to the end of the document. Useful for creating lists of theorems in the beginning of documents, before they've been stated. Overrides `thm-display.at`.

Default: `false`

thm-stored `state`

State containing theorem environment data, as an array of `thm` dictionaries.

A.3. thm-counter

Exposes methods for manipulating counters used by theorem environments. Import all functions using

```
#import thm-counter: *
```

- thm-counter-get()
- thm-counter-step()
- thm-counter-update()

Variables:

- thm-counters

thm-counter-get

Get theorem counter. Must be wrapped in context. See `thm.counter`.

Parameters

```
thm-counter-get(counter: str) -> array
```

thm-counter-step

Step theorem counter forward, relative to a `base` and `base-level`. Must be wrapped in context. See `thm.counter`, `thm.base`, `thm.base-level`.

Parameters

```
thm-counter-step(
  counter: str,
  base: str,
  base-level: int
)
```

thm-counter-update

Update theorem counter See `thm.counter`.

Parameters

```
thm-counter-update(
  counter: str,
```

```
  update: function
)
```

thm-counters `state`

State containing a dictionary of theorem environment counters. Each dictionary key is a counter name, and each entry is a counter value (array of integers).

B. Themes

B.1. thm-themes.ams

This theme defines the following versions of `thm` in the AMS style.

- `thm` in the `plain` style, intended for theorems, lemmas, corollaries, propositions, conjectures.
- `thm-def` in the `definition` style, intended for definitions, conditions, problems, examples.
- `thm-rem` in the `remark` style, intended for remarks, notes, annotations, claims, cases, acknowledgments, conclusions.

These styles have been used to provide the following environments.

- `theorem`, `proposition`, `lemma`, `conjecture`, `definition` sharing the counter `"Theorem"`, with the default `base "heading"`.
- `corollary`, `example` sharing the counter `"Sub-Theorem"`, with `base "Thoerem"`.
- `problem` with its own counter `"Problem"`, with the default `base`.
- `remark`, `claim`, `proof` (un-numbered).

```
#import "@preview/ctheorems:2.0.0": *
#import thm-themes.ams: *
#show: thm-rules

#set heading(numbering: "1.")

= Convergence in Probability

#definition[
  We say that $\{X_n\}$ converges to $X$ in
probability if for every $epsilon > 0$, $
    PP(|X_n - X| > epsilon) -> 0
  $ as $n -> oo$.
]
#remark[
  This is denoted $X_n ->^p X$.
]

#example[
  Let $EE[X_n] = 0$ and $EE[X_n^2] -> 0$.
  Then, $X_n ->^p 0$.
]

#theorem[Weak Law of Large Numbers][
  Let $\{X_n\}$ be i.i.d. with $EE[X_1] = mu$,
$E[X_1^2] < oo$. Then $
    1/n sum_(i = 1)^n X_i ->^p mu.
  $
]
```

1. Convergence in Probability

Definition 1.1. We say that $\{X_n\}$ converges to X in probability if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0$$

as $n \to \infty$.

Remark. This is denoted $X_n \xrightarrow{p} X$.

Example 1.1.1. Let $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] \to 0$. Then, $X_n \xrightarrow{p} 0$.

Theorem 1.2 (Weak Law of Large Numbers)**.** *Let $\{X_n\}$ be i.i.d. with $\mathbb{E}[X_1] = \mu$, $E[X_1^2] < \infty$. Then*

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu.$$

B.2. thm-themes.stripe

This theme creates theorem styles and environments just like `thm-themes.ams`, but with a formatting function `thm-fmt-stripe` which produces colorful stripes on the left sides of theorems.

```
#import "@preview/ctheorems:2.0.0": *
#import thm-themes.stripe: *
#show: thm-rules

#set heading(numbering: "1.")

= Convergence in Probability

#definition[
  We say that $\{X_n\}$ converges to $X$ in
probability if for every $epsilon > 0$, $
    PP(|X_n - X| > epsilon) -> 0
  $ as $n -> oo$.
]
#remark[
  This is denoted $X_n ->^p X$.
]

#example[
  Let $EE[X_n] = 0$ and $EE[X_n^2] -> 0$.
  Then, $X_n ->^p 0$.
]

#theorem[Weak Law of Large Numbers][
  Let $\{X_n\}$ be i.i.d. with $EE[X_1] = mu$,
$E[X_1^2] < oo$. Then $
    1/n sum_(i = 1)^n X_i ->^p mu.
  $
]

#proof[
  #lorem(14)
]
```

1. Convergence in Probability

Definition 1.3. We say that $\{X_n\}$ converges to X in probability if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0$$

as $n \to \infty$.

Remark. This is denoted $X_n \xrightarrow{p} X$.

Example 1.3.1. Let $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] \to 0$. Then, $X_n \xrightarrow{p} 0$.

Theorem 1.4 (Weak Law of Large Numbers). *Let $\{X_n\}$ be i.i.d. with $\mathbb{E}[X_1] = \mu$, $E[X_1^2] < \infty$. Then*

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu.$$

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut. ■

thm-fmt-stripe

Theorem formatting function, producing a colorful stripe on the left side. Wrapper around `thm-fmt-block()`.

Parameters

```
thm-fmt-stripe(
  thm: dictionary,
  ..args: arguments,
  stripe: stroke none,
  inset: relative dictionary
) -> content
```

..args `arguments`

Arguments, passed on to `thm-fmt-block()`

stripe `stroke` or `none`

Stroke of the stripe.

Default: `2pt + black`

inset `relative` or `dictionary`

Inset for the block.

Default: `(x: 1.0em, y: 0.2em)`